CSE4001 UNIVERSITY CURRICULUM SYNCHRONIZATION

Web Development Master Plan: Start to Jun 2027

Start: Nov 09, 2026 | Syllabus-Sync: CAT-1, 2 & TEE | Zero-Task Exam Lockouts

> **Strategic Architecture:** This roadmap aligns your Chapters 11 through 68 directly with the **CSE4001 (Internet and Web Programming)** university syllabus. While full-stack completion has no strict deadline, all syllabus-aligned topics are scheduled to finish *on or before* their respective examination milestones (**CAT-1: Units 1 & 2**, **CAT-2: Units 3 & 4**, and **TEE: Unit 5**). Zero-task lockouts ensure your 9.25+ CGPA remains shielded.

PHASE 1: PRE-CAT-1 SYNC — CSS3, RESPONSIVE DESIGN & BOOTSTRAP (NOV 09 – DEC 12, 2026)

Directly covers **CSE4001 Unit 2 (HTML5 & CSS3 Basics, Selectors, Box Model, Text Effects, Layouts, Media)** and Lab Tasks 1–3.

CALENDAR WINDOW	DAYS	CHAPTER & TOPIC FOCUS	LEVEL	CSE4001 SYLLABUS & LAB ALIGNMENT
Nov 09 – Nov 16	**8 Days** (6 Learn + 2 Rev)	**11-12. CSS (Parts 5 & 6)**	Medium	Flexbox containers, Grid systems, alignment, CSS transitions, transforms, media queries, CSS3 selectors (Unit 2).
Nov 17 – Nov 22	**6 Days** (4 Build + 2 Polish)	**13. Mini Project (CSS)**	Medium	Responsive landing page, Periodic table layout (Lab Task 3), image mapping styling (Lab Task 2).
Nov 23 – Nov 30	**8 Days** (6 Learn + 2 Rev)	**14. Bootstrap Framework**	Easy-Med	Grid system, responsive utilities, navbar, modal, forms, cards, spacing classes (Preps responsive Lab Tasks).
Dec 01 – Dec 12	**12 Days** (9 Build + 3 Rev)	**15. Comprehensive CSS Project**	Medium	Full multi-page responsive web layout with custom CSS & Bootstrap. Consolidates 100% of CSE4001 Unit 2.
Dec 13 – Dec 23	**11 Days**	**CAT-1 BUFFER & EXAM WINDOW (LOCKED): 0 Web Dev Tasks | 100% University Exam Prep & GPA Shield**		
Dec 24 – Dec 25	**2 Days**	**Post-CAT-1 Rest & Christmas Holiday: Complete rest, physical reset, zero coding.**		

PHASE 2: PRE-CAT-2 SYNC — COMPLETE JAVASCRIPT, DOM, EVENTS & AJAX (DEC 26 – FEB 02, 2027)

Directly covers **CSE4001 Unit 3 (JS, DOM, Events, Regex) & Unit 4 (JSON, AJAX basics)**, covering Lab Tasks 4–7.

CALENDAR WINDOW	DAYS	CHAPTER & TOPIC FOCUS	LEVEL	CSE4001 SYLLABUS & LAB ALIGNMENT
Dec 26 – Jan 01	**7 Days** (5 Learn + 2 Rev)	**16-19. JS Core (Parts 1–4)**	Easy-Med	Variables, datatypes, operators, conditionals, loops, array methods, string methods, functions (Unit 3).
Jan 02 – Jan 08	**7 Days** (5 Learn + 2 Rev)	**20-23. JS Advanced (Parts 5–8)**	Medium	Scopes, closures, `this`, arrow functions, array iteration (map/filter/reduce), spread/rest, destructuring.
Jan 09 – Jan 16	**8 Days** (4 Learn + 4 Break)	**24-25. DOM & Events (Parts 9–10)**	Med-Hard	DOM tree selection, manipulation, event listeners, bubbling, form events, Tic-Tac-Toe (Lab 5), Matrix (Lab 6).
Jan 17 – Jan 21	**5 Days** (3 Build + 2 Rev)	**26. JS Mini-Project (DOM Game)**	Medium	Interactive JS application (e.g., Simon Says / Tic-Tac-Toe / Hospital Form validator for Lab Task 7).
Jan 22 – Jan 27	**6 Days** (4 Learn + 2 Rev)	**27-29. Async JS, APIs & Regex (11–13)**	Hard	Call stack, Promises, `async/await`, Fetch API, Axios, JSON, Regex validation (ISBN validator Lab Task 4).
Jan 28 – Feb 02	**6 Days** (4 Learn + 2 Rev)	**30-31. Terminal & Git/GitHub**	Easy-Med	CLI navigation, flags, package managers, `git init`, branch workflows, conflict resolution, GitHub repos.
Feb 03 – Feb 13	**11 Days**	**CAT-2 BUFFER & EXAM WINDOW (LOCKED): 0 Web Dev Tasks | 100% University Exam Prep & GPA Shield**		

Milestone by Feb 02: All frontend styling, responsive design, JavaScript programming, DOM manipulation, Regex validation, and API consumption are complete, covering 100% of CAT-1 and CAT-2 requirements.

PRE-TEE BACKEND SYNC & POST-TEE SUMMER EXTENSION

Backend, SQL, Full-Stack MVC & React Ecosystem

Pre-TEE Sync: Mar 10 | TEE Lock: Mar 11–31 | Summer Track: Apr–Jun

PHASE 3: PRE-TEE SYNC — NODE, EXPRESS, REST & SQL DATABASE (FEB 14 – MAR 10, 2027)

Directly covers **CSE4001 Unit 5 (Server-side architecture, Form Handling, Sessions, Cookies & SQL Database Integration)**.

CALENDAR WINDOW	DAYS	CHAPTER & TOPIC FOCUS	LEVEL	CSE4001 SYLLABUS & LAB ALIGNMENT
Feb 14 – Feb 19	**6 Days** (4 Learn + 2 Rev)	**32-33. Node.js & Express.js**	Medium	Node runtime, `require` vs `import`, `fs` module, Express server, routing, params, query strings (Unit 5 sync).
Feb 20 – Feb 26	**7 Days** (5 Learn + 2 Rev)	**34-36. EJS, Misc & RESTful APIs**	Hard	Dynamic views, partials, GET vs POST body parsing, 7 standard REST routes, `method-override` for PUT/DELETE.
Feb 27 – Mar 05	**7 Days** (5 Learn + 2 Rev)	**37-38. SQL (Parts 1 & 2)**	Medium	Relational databases, tables, datatypes, primary/foreign keys, joins, aggregate queries (Unit 5 DB sync).
Mar 06 – Mar 10	**5 Days** (4 Learn + 1 Rev)	**39. Backend 5 (Node with SQL)**	Medium	Connecting Node/Express backend to relational SQL database via `mysql2`, connection pooling, parameterized queries.
Mar 11 – Mar 31	**21 Days**	**TEE / UNIVERSITY FINAL EXAM WINDOW: 0 Web Dev Tasks | 100% University Exam Immersion & GPA Shield**		

Curriculum Synchronization Checkpoint (March 10, 2027): 100% of CSE4001 (HTML5, CSS3, JavaScript, DOM, Events, Regex, JSON, AJAX, Node.js, Express, REST, and SQL) is completed prior to the TEE window, ensuring you enter the final theory and lab FAT exams prepared.

PHASE 4: SUMMER TRACK — NOSQL, FULL-STACK PRODUCTION & REACT (APR 01 – JUN 01, 2027)

Completed during the post-TEE semester break and summer term with open pacing for internship and portfolio preparation.

CALENDAR WINDOW	DAYS	CHAPTER & TOPIC FOCUS	LEVEL	INDUSTRY PRODUCTION DELIVERABLES
Apr 01 – Apr 08	**8 Days**	**40-43. MongoDB & Mongoose**	Medium	Mongo installation, `mongosh`, collections, Mongoose schemas, models, CRUD operations with Express.
Apr 09 – Apr 17	**9 Days**	**44-48. Project Phase 1 + Middleware**	Hard	Major Project CRUD listings, EJS-mate layouts, custom middleware (`next()`), centralized error handling.
Apr 18 – Apr 27	**10 Days**	**49-54. Relationships & Auth (Phase 2)**	Hard	Mongoose `populate()`, cascade deletes, Joi validations, cookies, sessions, Passport.js authentication.
Apr 28 – May 05	**8 Days**	**55-58. MVC, Cloudinary & Security**	Very Hard	MVC controller refactor, Cloudinary image uploads (Multer), Mapbox geocoding, Helmet security headers, deployment.
May 06 – May 18	**13 Days**	**59-65. Modern React Ecosystem**	Hard	Vite build, JSX, components, props, `useState`, complex state, controlled forms, `useEffect`, React Router v6.
May 19 – May 27	**9 Days**	**66. Full MERN Capstone Project**	Very Hard	Decoupled full-stack React SPA connected to Express/Mongo API with complete authentication & state handling.
May 28 – Jun 01	**5 Days**	**67-68. Tailwind CSS & Redux Toolkit**	Medium	Utility-first responsive layouts with Tailwind CSS, global state management slices, reducers, and RTK hooks.